

# Supplemental Material for “Interband Pairing Signatures in the Superfluid Response of Magic-Angle Twisted Bilayer Graphene”

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TABLE I. Default parameters used for the mechanism scans.

| quantity                   | value                                                      |
|----------------------------|------------------------------------------------------------|
| twist angle                | $\theta = 1.05^\circ$                                      |
| Dirac velocity             | $v_F = 2.135 \text{ eV \AA}$                               |
| AA tunneling               | $w_0 = 87.2 \text{ meV}$                                   |
| AB tunneling               | $w_1 = 109.0 \text{ meV}$                                  |
| reciprocal cutoff          | $N_{\text{shell}} = 3$                                     |
| retained bands             | $n_{\text{keep}} = 6$                                      |
| momentum grid              | $nk = 7$ for dense maps, $nk = 9$ for key-point validation |
| gap scale                  | $\Delta_0 = 1 \text{ meV}$                                 |
| velocity finite difference | $dk = 10^{-6} \text{ \AA}^{-1}$                            |

## I. SCOPE OF THE SUPPLEMENTAL MATERIAL

This Supplemental Material documents the executable numerical workflow behind the PRB manuscript. It is organized as a reproducibility and self-audit record: the Bistritzer-MacDonald (BM) model conventions [1], the orbital-projected pairing construction, the finite-band response formula, the figure and table provenance, and the historical PRB benchmark reconstruction are recorded separately.

The main text makes a conservative mechanism claim: an orbital-projected interband-pairing direction produces a reproducible, normalization-conditioned change in the normalized finite-band superfluid response. The present Supplemental Material does not promote the raw response units to final experimental stiffness values. This choice is motivated by the distinction between robust quantum-geometric mechanism diagnostics [2–5] and direct quantitative comparison with recent stiffness experiments [6, 7].

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## II. BM MODEL AND NUMERICAL PARAMETERS

The normal-state calculation uses the single-valley BM continuum Hamiltonian with orbital basis

$$(tA, tB, bA, bB), \quad (1)$$

where  $t, b$  label layers and  $A, B$  label sublattices. Unless explicitly stated otherwise, the production mechanism scans use the parameters in Table I.

The moire unit-cell area and reciprocal scale used in the unit audit are

$$A_M = 1.5606 \times 10^4 \text{ \AA}^2, \quad G_M = 0.054047 \text{ \AA}^{-1}. \quad (2)$$

The raw response conversion used in the audit is

$$D[\text{eV \AA}^2] = \frac{D_{\text{raw}}[\text{meV \AA}^2]}{1000}, \quad (3)$$

with all remaining spin, valley, or convention factors kept explicit rather than absorbed into a hidden prefactor.

## III. ORBITAL-PROJECTED PAIRING CONSTRUCTION

The pairing direction is specified first in the layer-sublattice orbital basis,

$$\Delta_{\text{orb}}(\eta) = \Delta_0 \mathcal{N}_\eta (M_0 + \eta M_1), \quad (4)$$

with

$$M_0 = \tau_0 \sigma_0, \quad M_1 = \tau_x \sigma_x. \quad (5)$$

The two normalization controls used in the manuscript are

$$\text{fixed Frobenius:} \quad \|M_0 + \eta M_1\|_F = \|M_0\|_F, \quad (6)$$

$$\text{fixed } \Delta_0 : \quad \mathcal{N}_\eta = 1. \quad (7)$$

The band-basis pairing matrix is

$$\Delta_{\text{band}}(\mathbf{k}) = U^\dagger(\mathbf{k}) \Delta_{\text{orb}} U^*(-\mathbf{k}). \quad (8)$$

The working valley convention in the present diagnostic is the time-reversal sewn proxy

$$U_-(-\mathbf{k}) = U_+^*(\mathbf{k}). \quad (9)$$

This convention is useful for isolating an orbital-defined interband-pairing signature, but the readiness audit identifies a fully documented two-valley implementation as a submission-critical improvement.

The projected interband-pairing weight is

$$W_{\text{inter}} = \frac{\sum_{n \neq m} |\Delta_{nm}|^2}{\sum_{n,m} |\Delta_{nm}|^2}. \quad (10)$$

For the dense  $nk = 7$ ,  $n_{\text{keep}} = 6$  scan,  $W_{\text{inter}} \simeq 0.108$  at  $\eta = 1$ .

#### IV. FINITE-BAND RESPONSE AND CHANNEL SPLIT

For a retained band subspace, the finite-band BdG Hamiltonian is

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) - \mu & \Delta_{\text{band}}(\mathbf{k}) \\ \Delta_{\text{band}}^\dagger(\mathbf{k}) & -[\varepsilon(\mathbf{k}) - \mu] \end{pmatrix}. \quad (11)$$

The zero-temperature static current-current response is evaluated as

$$\Lambda_{\alpha\beta} = \sum_{a \neq b} \frac{\langle a | J_\alpha | b \rangle \langle b | J_\beta | a \rangle (f_a - f_b)}{E_b - E_a}. \quad (12)$$

The band-basis velocity is split into diagonal and off-diagonal parts,

$$v_\alpha^{\text{band}} = v_\alpha^{\text{intra}} + v_\alpha^{\text{inter}}. \quad (13)$$

This produces conventional, geometric, and cross-channel response diagnostics. The primary normalized response map uses

$$\delta D_{\text{iso}}(\mu, \eta) = \frac{D_{\text{iso}}(\mu, \eta)}{D_{\text{iso}}(\mu, 0)} - 1, \quad D_{\text{iso}} = \frac{D_{xx} + D_{yy}}{2}. \quad (14)$$

#### V. FIGURE AND TABLE PROVENANCE

Table II summarizes the current figure and table provenance. The full machine-readable record is maintained in `notes/figure_table_provenance_2026-06-09.md`.

The dense response data are reproduced by

```
python3 scripts/run_mu_response_scan.py \
--nk 7 --n-shell 3 --n-keep 6 \
```

TABLE II. Manuscript artifact provenance. CSV values used in the manuscript tables are checked by `scripts/verify_manuscript_tables.py`.

| artifact                  | data source           | script          | check              |
|---------------------------|-----------------------|-----------------|--------------------|
| Frobenius heatmap         | dense scan CSV        | heatmap plotter | figure file exists |
| fixed- $\Delta_0$ heatmap | dense scan CSV        | heatmap plotter | figure file exists |
| dense response table      | eta-summary CSV       | eta summarizer  | table verifier     |
| $nk = 9$ validation table | key-point summary CSV | eta summarizer  | table verifier     |
| PRB audit table           | reconstruction CSVs   | audit scripts   | table verifier     |

```
--mus -5 -4 -3 -2 -1 0 1 2 3 4 5 \
--etas 0 0.25 0.5 0.75 1 \
--output data/processed/mu_eta_response_scan_nk7_nkeep6.csv
```

and summarized by

```
python3 scripts/summarize_eta_effect.py \
data/processed/mu_eta_response_scan_nk7_nkeep6.csv \
--output data/processed/mu_eta_response_scan_nk7_nkeep6_eta1_summary.csv
```

The table verifier is

```
python3 scripts/verify_manuscript_tables.py
```

## VI. HISTORICAL PRB BENCHMARK RECONSTRUCTION

The production interband-pairing scan and the historical PRB table reconstruction are kept separate. The best simple unified reconstruction route for the old uniform- $s$  benchmark is

$$D_{\text{conv}} = 2D_{\text{intra},\tau_z}, \quad D_{\text{geom}} = D_{\text{inter},\tau_z}. \quad (15)$$

For  $n_{\text{keep}} = 6$ , this gives  $D_{\text{total}} = 126.66 \text{ eV } \text{\AA}^2$  versus the historical target  $129.30 \text{ eV } \text{\AA}^2$ , and  $D_{\text{geom}} = 75.22 \text{ eV } \text{\AA}^2$  versus  $75.30 \text{ eV } \text{\AA}^2$ .

The old  $n_{\text{keep}} = 2$  endpoint is more convention-sensitive. The closest audited reconstruction combines a Gamma-centered mesh, central-pair band selection, an all- $\tau_z$  current

convention, a factor of two in the intraband term, and a full curvature-like conventional correction. This gives

$$D_{\text{total}} = 66.18 \text{ eV } \text{\AA}^2 \quad \text{versus} \quad 67.50 \text{ eV } \text{\AA}^2. \quad (16)$$

Because this full-curvature correction does not work uniformly at  $n_{\text{keep}} = 6$ , the reconstruction is treated as a historical audit rather than a production convention.

## VII. CURRENT LIMITATIONS AND NEXT VALIDATION GATES

The current submission-critical gaps are:

1. a higher-accuracy convergence check beyond the present  $nk = 9$  key points, preferably at  $nk = 11$  for selected  $\mu$  values;
2. a fully documented two-valley sewing convention or an explicitly limited diagnostic-model framing;
3. a final absolute stiffness convention before any direct experimental comparison;
4. a calibrated filling variable replacing the retained-band proxy  $\nu_{\text{proxy}}$ .

These limitations do not invalidate the normalized-response mechanism claim, but they prevent the current draft from being treated as a final PRB submission without further convergence and convention work.

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## DECLARATIONS

**Funding:** Not applicable.

**Conflict of interest:** The author declares no conflict of interest.

**Data availability:** All data and code supporting this work are available from the corresponding author upon reasonable request.

**Ethics approval:** Not applicable.

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